

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Inferences	Hard

Question

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps’ burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow’s measurements both at a moment in time and throughout the entire burrow-building process, something that’s impossible with resin casting because _____

Which choice most logically completes the text?

Answer

- A. it can only be used on burrows below a certain size.
- B. it does not allow for multiple castings of the same burrow over time.
- C. the casting process takes more time than *A. bellulus* takes to construct a burrow.
- D. the process of recovering the model distorts the resin’s shape.

Correct Answer: B

Rationale

Choice B is the best answer. Since resin casting “inevitably requires destroying the burrow,” it would be impossible to make multiple castings of the same burrow over time.

Choice A is incorrect. The passage doesn’t discuss size requirements for completing resin casting, so there’s no basis for this inference. Choice C is incorrect. The passage never mentions how long the casting process takes nor how long *A. bellulus* takes to construct a borrow, so there’s no basis for this inference. Choice D is incorrect. The passage never states that recovering the model distorts the resin’s shape, only that it destroys the burrow. Therefore, there’s no basis for this inference.